

HireCraft AI: Core Module Description

1. User Authentication & Profiling Module

- **Purpose:** Manages secure candidate onboarding, profile personalization, and resume metadata storage.
- **Functionality:** Handles user registration, login authentication, target role selection, and PDF resume processing.
- **Tables Involved:** users
- **Data & Workflow:** Stores BCrypt encrypted credentials, account states, academic metrics (CGPA, department), and JSONB-structured AI resume analysis alongside object storage links.

2. Virtual Interview (Mock Assessment Center) Module

- **Purpose:** Orchestrates complete, end-to-end multi-stage placement drives mimicking real company recruitment processes.
- **Functionality:** Guides candidates sequentially through the fixed workflow: Aptitude → Hard DSA → Technical → Behavioral → Evaluation Report.
- **Tables Involved:** virtual_interviews, alongside optional foreign keys linking coding_assessments, aptitude_assessments, and interview_sessions.
- **Data & Workflow:** Tracks overall session progression (IN_PROGRESS or COMPLETED), current stage execution, and computes weighted cumulative final scores.

3. Coding Assessment & DSA Execution Module

- **Purpose:** Evaluates algorithmic problem-solving capabilities using LeetCode-style challenges within a secure execution sandbox.
- **Functionality:** Fetches topic-wise or company-wise coding problems, tracks timer constraints, prevents copy-pasting, and executes candidate source code.
- **Tables Involved:** coding_assessments, coding_submissions
- **Data & Workflow:** Records assessment metadata (mode, difficulty) in coding_assessments and links multiple candidate attempts, raw source code, and direct Judge0 sandbox metrics (stdout, stderr, execution time, and test case pass counts) into coding_submissions.

4. Aptitude Assessment Module

- **Purpose:** Evaluates quantitative, logical, and verbal reasoning skills through timed MCQ assessments.
- **Functionality:** Presents question sets, tracks question-wise time consumption, evaluates selected choices, and identifies weak conceptual areas.
- **Tables Involved:** aptitude_assessments, aptitude_answers
- **Data & Workflow:** Stores quiz session aggregates (score, accuracy) in aptitude_assessments and records individual candidate-selected options (A, B, C, D), correctness flags, and timing metrics in aptitude_answers.

5. Technical & Behavioral Interview Module (Text & WebRTC Voice)

- **Purpose:** Conducts conversational technical (core CS, project, resume) and behavioral (STAR-format, situational) interviews.
- **Functionality:** Integrates with LiveKit WebRTC rooms and Groq Whisper speech-to-text (STT) services to capture spoken answers and evaluate communication metrics.
- **Tables Involved:** interview_sessions, interview_answers
- **Data & Workflow:** Manages session initialization and LiveKit room identifiers in interview_sessions while capturing per-question text inputs, STT transcripts, response latency, and duration logs in interview_answers.

6. Unified AI Evaluation & Reporting Module

- **Purpose:** Consolidates structured AI feedback, automated scoring, and comprehensive markdown performance reviews.
- **Functionality:** Compiles evaluation results across resume analysis, coding rounds, and interview sessions into unified feedback documents.
- **Tables Involved:** evaluations, reports
- **Data & Workflow:** Uses strict Exclusive Arc foreign key constraints to securely link evaluations and reports to specific assessments without orphaned records, storing structured JSONB strengths/weaknesses and markdown content.

7. Central AI Memory & Readiness Analytics Module

- **Purpose:** Powers platform-wide personalization by learning candidate patterns and tracking historical readiness progression.
- **Functionality:** Dynamically updates long-term behavioral facts, weak concepts, and multi-dimensional placement probabilities after every test attempt.

- **Tables Involved:** ai_memory_items, readiness_snapshots
- **Data & Workflow:** Stores active/resolved AI memory attributes in ai_memory_items and creates immutable time-series records of resume, coding, aptitude, and overall readiness scores in readiness_snapshots for dashboard radar charts.